

Problem

A professional center is supplied by a balanced three-phase source. The center has four balanced three-phase loads as follows:

Load 1: 150 kVA at 0.8 pf leading

Load 2: 100 kW at unity pf

Load 3: 200 kVA at 0.6 pf lagging

Load 4: 80 kW and 95 kVAR (inductive)

If the line impedance is $0.02 + j0.05 \Omega$ per phase and the line voltage at the loads is 480 V, find the magnitude of the line voltage at the source.

Step-by-step solution

Step 1 of 2

$$P f = 0.8(\text{Leading}) \Rightarrow \theta_1 = -36.87^\circ$$

$$S_1 = 150 \angle -36.87^\circ \text{ KVA}$$

$$P f = 1.0 \Rightarrow \theta_2 = 0^\circ$$

$$S_2 = 100 \angle 0^\circ \text{ KVA}$$

$$P f = 0.6 (\text{lagging}) \Rightarrow \theta_3 = 53.13^\circ$$

$$S_3 = 200 \angle 53.13^\circ \text{ KVA}$$

$$S_4 = 80 + j95 \text{ KVA}$$

$$S = S_1 + S_2 + S_3 + S_4 = 420 + j165 = 451.2 \angle 21.45^\circ \text{ KVA}$$

$$S = \sqrt{3} V_L I_L$$

$$\Rightarrow I_L = \frac{S}{\sqrt{3} V_L} = \frac{451.2 \times 10^3}{\sqrt{3} \times 480} = 542.7 \text{ A}$$

Step 2 of 2

For the line,

$$S_L = 3 I_L^2 Z_L = 3(542.7)^2 (0.02 + j0.05)$$

$$S_L = 17.67 + j44.18 \text{ KVA}$$

At the source,

$$S_T = S + S_L = 437.7 + j209.2$$

$$= 485.1 \angle 25.55^\circ \text{ KVA}$$

$$V_T = \frac{S_T}{\sqrt{3} I_L} = \frac{485.1 \times 10^3}{\sqrt{3} \times 542.7} = 516 \text{ V}.$$